

Assignment-8

1. Write a Python program to accept **name and age from the user** and check whether the person is **eligible for voting**.
2. Write a Python program to perform **arithmetic operations** on two numbers.

Operations:

- Addition
 - Subtraction
 - Multiplication
 - Division
3. Write a Python program using a **function** to calculate the **square and cube of a number**.
 4. Write a Python program to **store 5 numbers in a list** and print their **sum and average**.
 5. Write a Python program to **find the maximum and minimum number in a list**.

Example

```
Input  : [10,20,5,40,30]
Output : Max = 40
        Min = 5
```

6. Write a Python program to **remove duplicate values from a list using set**.

Example

```
Input  : [1,2,2,3,4,4,5]
Output : [1,2,3,4,5]
```

7. Write a Python program to **sort a list in ascending and descending order**.
8. Write a Python program to **filter even numbers from a list**.

Example

```
Input  : [10,13,58,90,24]
Output : [10,58,90,24]
```

9. Write a Python program to calculate **square of all numbers in a list using map()**.

Example

```
Input  : [1,2,3,4]
Output : [1,4,9,16]
```

10. Write a Python program to create a **dictionary storing student information**:

```
name
age
marks
```

Then display the dictionary.

11. Write a Python program to **check whether a key exists in a dictionary**.

12. Write a Python program to **reverse a string**.

Example

```
Input  : Python
Output : nohtyP
```

13. Write a Python program to **count occurrences of an element in a list**.

Example

```
Input  : [10,20,10,30,10]
Output : 10 appears 3 times
```

14. Write a Python program to **count the frequency of characters in a string using a dictionary**.

Example

```
Input  : python
Output :
p:1
y:1
t:1
h:1
o:1
n:1
```

15. Write a Python program to **create and print a 2D list (multidimensional list)**.

Example

```
[  
  [10,20],  
  [30,40],  
  [50,60]  
]
```

